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Student Number

2015

MID COURSE
EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time 3 hours
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-16 show relevant mathematical reasoning and/or calculations
- Start a new booklet for each question

Total Marks – 100

Section I - Pages 2 - 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II - Pages 6 - 12

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hour and 45 minutes for this section

Question	Marks
1 - 10	/10
11	/15
12	/15
13	/15
14	/15
15	/15
16	/15

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 30% of the Higher School Certificate Course Assessment

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1 – 10 (Detach from paper)

1 What is 5.03976 correct to 4 significant figures?

- (A) 5.039
- (B) 5.0397
- (C) 5.040
- (D) 5.0398

2 The function $y = 3 + (x - 2)^2$ is:

- (A) An EVEN function
- (B) An ODD function
- (C) Symmetric about $x = 2$
- (D) Symmetric about $y = 3$

3 The solution to $(2x - 5)(6 - x) \geq 0$ is:

- (A) $\{x : -2.5 \leq x \leq 6\}$
- (B) $\{x : 2.5 \leq x \leq 6\}$
- (C) $\{x : x \leq 2.5, x \geq -6\}$
- (D) $\{x : x \leq -2.5, x \geq 6\}$

Section 1 continues on page 3

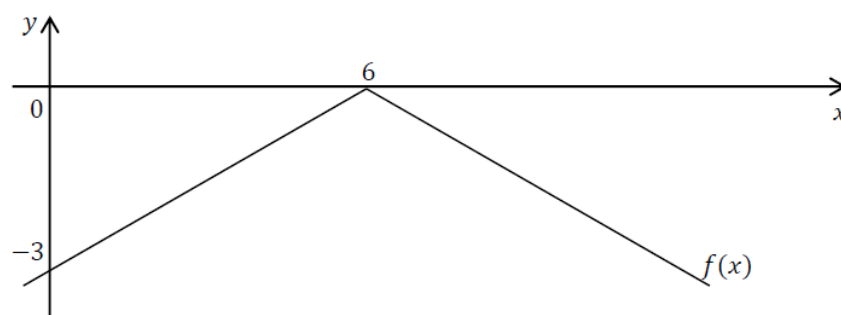
4 If $\int_0^a (4 - 2x) dx = 4$, the value of a is:

- (A) -2
- (B) 0
- (C) 1
- (D) 2

5 What is the value of $\int_{-2}^2 \sqrt{4 - x^2} dx$?

- (A) $\frac{3\pi}{2}$
- (B) 2π
- (C) 3π
- (D) 4π

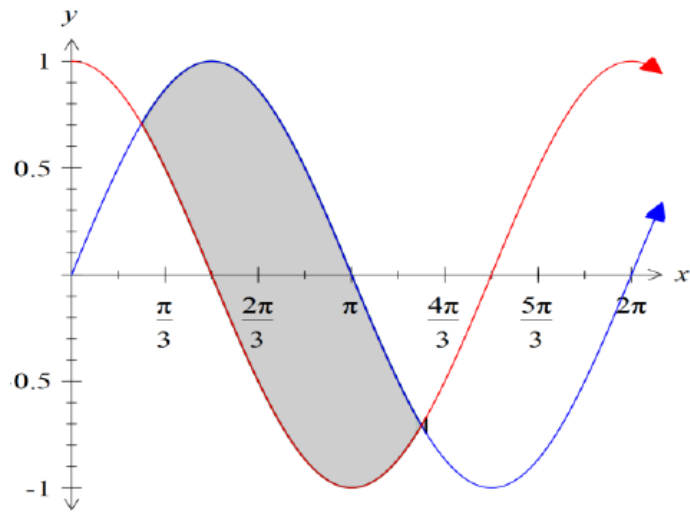
6



The diagram shows the graph of $y = f(x)$. The equation of $y = f(x)$ is:

- (A) $f(x) = \left| \frac{1}{2}x - 3 \right|$
- (B) $f(x) = -|2x - 3|$
- (C) $f(x) = -\left| \frac{1}{2}x - 3 \right|$
- (D) $f(x) = -\left| \frac{1}{2}x + 3 \right|$

Section 1 continues on page 4



Which of the following describes the area given in the graph above?

(A) $\int_{\frac{\pi}{3}}^{\frac{4\pi}{3}} (\sin x - \cos x) dx$

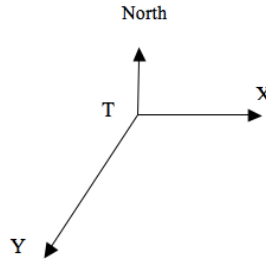
(B) $\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x - \cos x) dx$

(C) $\int_{\frac{\pi}{3}}^{\frac{4\pi}{3}} (\cos x - \sin x) dx$

(D) $\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\cos x - \sin x) dx$

Section 1 continues on page 5

- 8 A point X is due East of a tower T . From another point Y , the bearing of the tower T is 061° . The size of angle XTY is equal to:



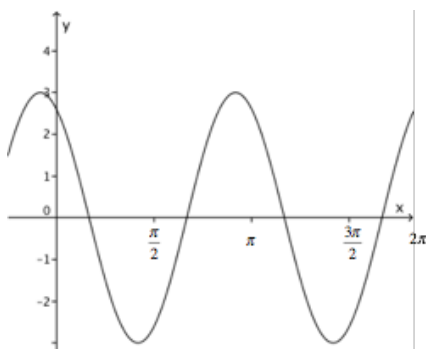
- (A) 131°
(B) 119°
(C) 209°
(D) 151°
- 9 What is the correct expression for $\int \cos \frac{x}{3} dx$?

- (A) $-\frac{1}{3}\sin \frac{x}{3} + C$
(B) $\frac{1}{3}\sin \frac{x}{3} + C$
(C) $-3\sin \frac{x}{3} + C$
(D) $3\sin \frac{x}{3} + C$

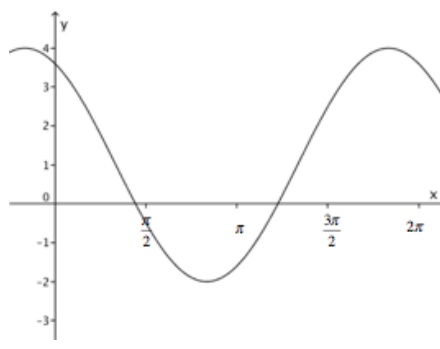
Section 1 continues on page 6

10 Which graph best represents $y = -3\sin\left(2x - \frac{\pi}{3}\right) + 1$

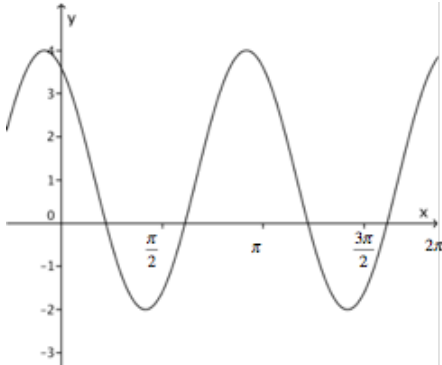
(A)



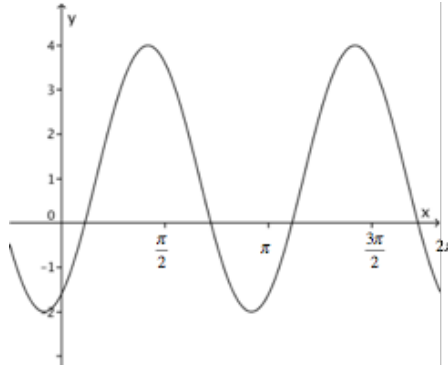
(B)



(C)



(D)



End of Section 1

Section II

90marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11(15 marks) Use a SEPARATE writing booklet

- (a) Factorise completely $4x^3 - 32$ 2
- (b) Solve $2x^2 - 7x - 15 = 0$ 2
- (c) Solve $|3x - 7| \leq 12$ 2
- (d) Simplify $\frac{2}{x-1} - \frac{1}{x-3}$ 2
- (e) Find values of a and b such that $\frac{-2}{7-\sqrt{3}} = a + b\sqrt{3}$ 3
- (f) Find the exact value of $\sin \frac{4\pi}{3}$ 2
- (g) State the domain and range of $y = \frac{1}{3-x}$ 2

End of Question 11

Question 12(15 marks) Use a SEPARATE writing booklet

(a) Differentiate the following:

(i) $(7 - 5x)^2$ 2

(ii) $\frac{2x}{\cos 2x}$ 3

(b) Find:

(i) $\int x^3 + 7x^2 + 4 \, dx$ 2

(ii) $\int \frac{x^3 + 1}{\sqrt{x}} \, dx$ 2

(iii) $\int_0^3 2\sqrt{x} + x^3 \, dx$ 2

(c) Given the parabola $4y = x^2 - 12$, find the:

(i) Focal length 1

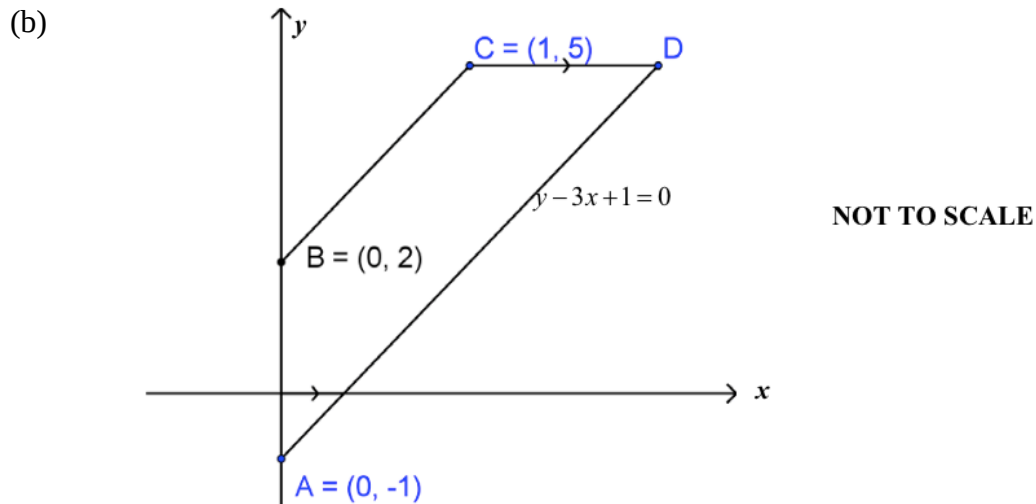
(ii) Coordinates of the focus 1

(d) Find the equation of the tangent to the curve $y = \frac{1}{2} \sin x$ at the point $(\pi, 0)$. 2

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) The roots of the quadratic $px^2 - x + q = 0$ are -2 and 5 . 2
Find the values of p and q .



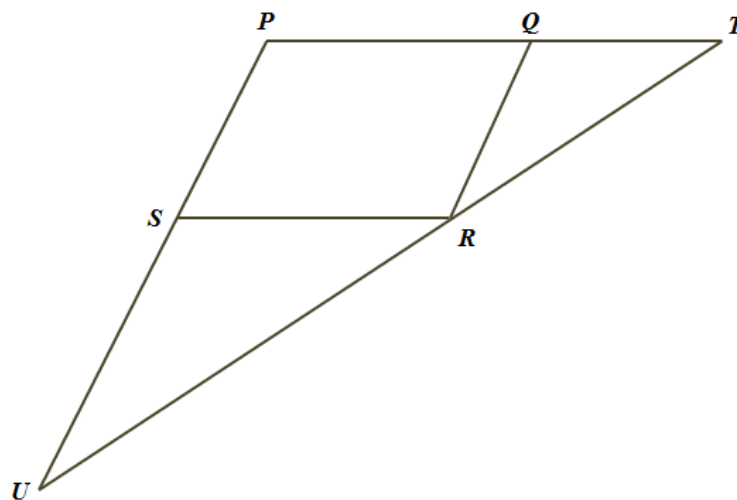
In the diagram, $ABCD$ is a quadrilateral. The equation of the line AD is $y - 3x + 1 = 0$

- (i) Show that $ABCD$ is a trapezium. 2
- (ii) The line CD is parallel to the x -axis. Find the coordinates of D . 1
- (iii) Find the length of BC . 1
- (iv) Show that the perpendicular distance from B to AD is $\frac{3}{\sqrt{10}}$ 2
- (v) Hence or otherwise, find the area of the trapezium $ABCD$. 2

Question 13 continues on Page 10

Question 13 continued

(c)



$PQRS$ is a parallelogram. PQ is produced to T so that $TQ = QR$ and PS is produced to U so that $SU = PS$. The points U , R and T are collinear.

(i) Prove that $\triangle TQR \equiv \triangle RSU$.

3

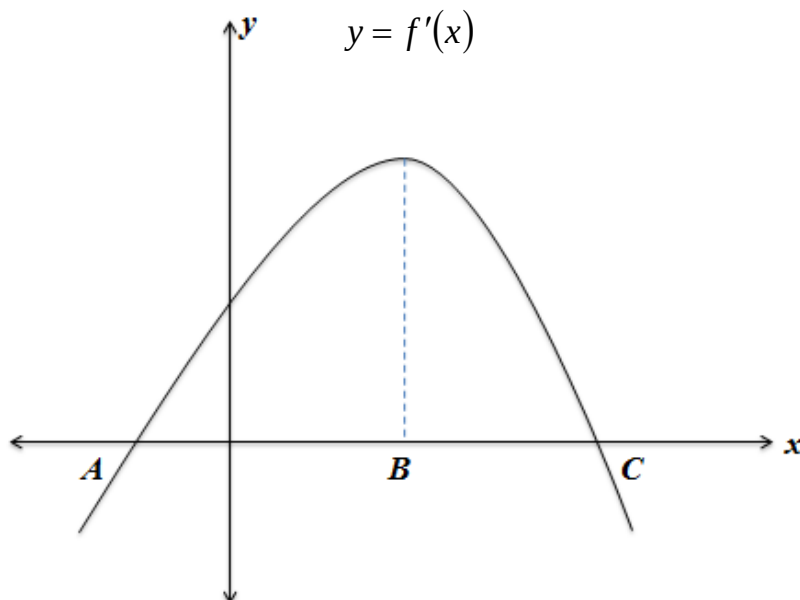
(ii) Hence, or otherwise, prove that $PQRS$ is a rhombus.

2

End of Question 13

Question 14(15 marks) Use a SEPARATE writing booklet

- (a) Given the curve $f(x)=2x^3-12x^2$:
- (i) Find the stationary points and establish their nature. 3
- (ii) Determine the points of inflection. 2
- (iii) Sketch the curve, in the domain $-1 \leq x \leq 6$, showing all the essential features. 3
- (b) The diagram shows the gradient function $y = f'(x)$ of a curve $y = f(x)$.

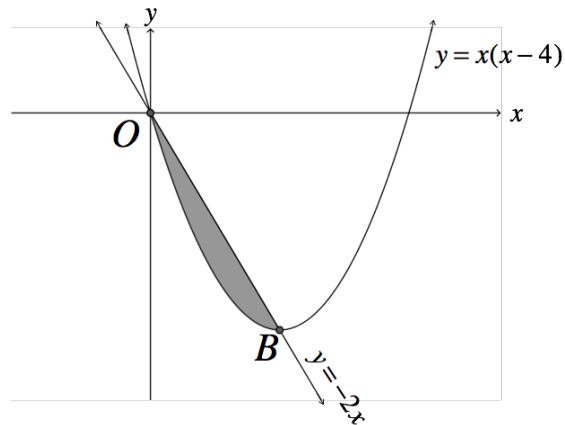


- (i) Describe the shape of the curve $y = f(x)$ where $x=A$ and $x=B$. 2
- (ii) For what values of x is the function $y = f(x)$ decreasing? 1
- (iii) Sketch a possible curve $y = f(x)$, showing where $x=A, B$ and C . 2
- (c) Evaluate $\int_{-1}^1 4^x dx$ using Simpson's rule with five function values. 2

End of Question 14

Question 15(15 marks) Use a SEPARATE writing booklet

(a)

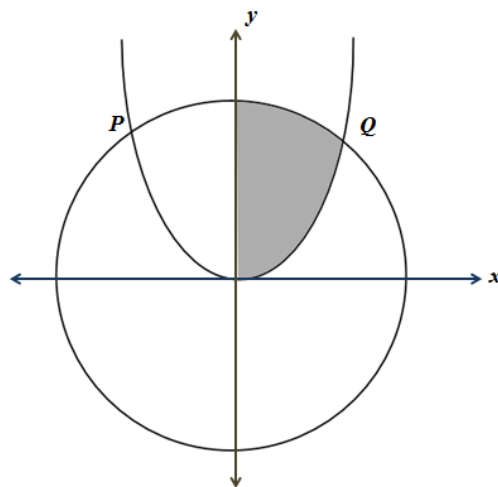


The graph of $y = x(x-4)$ and $y = -2x$ meet at $B(2, -4)$.

3

Find the shaded area bounded by $y = x(x-4)$ and $y = -2x$

(b) The shaded region is bounded by the curve $y = x^2$ and the circle $x^2 + y^2 = 12$.



(i) By solving the two equations simultaneously, show that the points P and Q both have a y -coordinate of 3.

2

(ii) Show that the volume generated when the shaded region is rotated about the y -axis is $\pi\left(16\sqrt{3} - \frac{45}{2}\right)$ cubic units.

3

Question 15 continues on page 13

Question 15 (continued)

- (c) Find the equation of the line which passes through the point of intersection of the lines $2y = x + 8$ and $6y = 4x + 27$ and has a gradient of $-\frac{3}{2}$. **3**

- (d) (i) Show that $\frac{d}{dx}(\sqrt{1+x^2}) = \frac{x}{\sqrt{1+x^2}}$ **1**

- (ii) Hence, evaluate $\int_0^1 \frac{x}{2\sqrt{1+x^2}} dx$ **3**

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet

(a) $\cos \alpha = -p$ and $\tan \alpha > 0$, find $\sec(90 + \alpha)$. 2

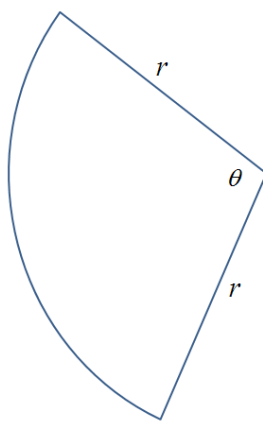
(b) (i) Show that $x = \frac{\pi}{3}, \frac{2\pi}{3}$ are the solutions of the equation 1
 $1 + 2 \cos 2x = 0$ for $0 \leq x \leq \pi$.

(ii) Draw the graph of $y = 1 + 2 \cos 2x$ for $0 \leq x \leq \pi$. 3

(iii) Find the area between $y = 1 + 2 \cos 2x$ and the x-axis for $\frac{\pi}{3} \leq x \leq \frac{2\pi}{3}$. 2

(c) A landscaping company is designing a garden bed in a local park in the shape of a sector with radius r and angle θ .

They need 375 metres of garden edging material to use in the perimeter of the garden bed.



(i) Show that the area of A of the garden bed is $\frac{r}{2}(375 - 2r)$. 2

(ii) Find the greatest area of the garden bed which can be made using 375 metres of edging material. 3

(iii) After inspecting the location for the garden bed, the designers calculate that the angle of the sector shaped garden bed will be less than 110° . 2

Can they still build the garden bed with the maximum area found in part (ii)? Justify your answer.

End of Paper

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Student Number

Mathematics

Section I – Multiple Choice Answer Sheet

Use this multiple-choice answer sheet for questions 1 – 10. Detach this sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 ↑ *correct*

**Start
Here**



1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2015 Half-yearly solutions

Multiple choice

1. A	2. C	3. B	4. D	5. B
6. C	7. B	8. D	9. D	10. C

Free response

Question 11

a)	$\frac{4x^3 - 32}{4(x^3 - 8)}$ $4(x - 2)(x^2 + 2x + 4)$	1 mark: takes 4 common factor out 1 mark: correctly factorises the difference of cubes	
b)	$2x^2 - 7x - 15 = 0$ $(2x + 3)(x - 5) = 0$ $x = \frac{-3}{2}, \quad x = 5$	1 mark: correctly factorises 1 mark: correctly gives the solution from the factored form.	
c)	$ 3x - 7 \leq 12$ $3x - 7 \leq 12 \quad - (3x - 7) \leq 12$ $3x \leq 19 \quad 7 - 3x \leq 12$ $x \leq \frac{19}{3} \quad -3x \leq 5$ $x \geq \frac{5}{3}$	1 mark for each correct answer with correct working.	
d)	$\frac{\frac{2}{x-1} - \frac{1}{x-3}}{\frac{2(x-3) - (x-1)}{(x-1)(x-3)}} \quad 1mark$ $\frac{2x - 6 - x + 1}{(x-1)(x-3)}$ $\frac{x - 5}{(x-1)(x-3)} \quad 1mark$	1 mark: makes it a single fraction with common denominator 1 mark: correctly simplifies	
e)	$\frac{-2}{7 - \sqrt{3}} = \frac{-2 \times (7 + \sqrt{3})}{(7 - \sqrt{3})(7 + \sqrt{3})}$ $= \frac{-14 - 2\sqrt{3}}{46}$ $= -\frac{7}{23} - \frac{1}{23}\sqrt{3}$	2 marks correct values of a and b using suitable working 1 mark – identifying the need to multiply by the conjugate and showing appropriate working	

	$\therefore a = -\frac{7}{23} \quad b = -\frac{1}{23}$		
f)	$\sin \frac{4\pi}{3}$ $\sin\left(\pi + \frac{\pi}{3}\right)$ $-\sin\left(\frac{\pi}{3}\right)$ $= \frac{-\sqrt{3}}{2}$	2 mark: correct answer from correct working 1 mark: correct bald answer given	
g)	$y = \frac{1}{3-x}$ Domain x all real except $x \neq 3$ Range y all real except $y \neq 0$	2 marks both range and domain correct 1 mark only one of range or domain correct.	

Question 12

a)(i)	$y = (7 - 5x)^2$ $\frac{dy}{dx} = 2(7 - 5x) \times -5$ $= -10(7 - 5x)$	2 mark: correctly differentiates 1 mark: gives $2(7 - 5x)$, forgets to multiply by the derivative of the bracket	
a)(ii)	$y = \frac{2x}{\cos 2x}$ $u = 2x \quad u' = 2$ $v = \cos 2x \quad v' = -2 \sin 2x$ $\frac{dy}{dx} = \frac{2 \cos 2x - -4x \sin 2x}{\cos^2 2x}$ $= \frac{2 \cos 2x + 4x \sin 2x}{\cos^2 2x}$	2 marks: correct answer 1 mark: attempts to use quotient rule from incorrect u' or v'	
b)(i)	$\int x^3 + 7x^2 + 4 dx$ $\frac{x^4}{4} + \frac{7x^3}{3} + 4x + C$	2 marks: correct answer 1 mark: attempts to integrate, but +C required	
b)(ii)	$\int \frac{x^3 + 1}{\sqrt{x}} dx$ $\int x^{\frac{5}{2}} + x^{\frac{-1}{2}} dx$ $\frac{2}{7} x^{\frac{7}{2}} + 2x^{\frac{1}{2}} + C$	1 mark: writes the integrand in the correct radical form 1 mark: correctly integrates	
b)(iii)	$\int_0^3 2\sqrt{x} + x^3 dx$ $\int_0^3 2x^{\frac{1}{2}} + x^3 dx$ $\left[\frac{4}{3} x^{\frac{3}{2}} + \frac{x^4}{4} \right]_0^3$ $\left(\frac{4}{3} \cdot 3\sqrt{3} + \frac{81}{4} \right) - 0$ $4\sqrt{3} + \frac{81}{4}$	1 mark: correctly integrates 1 mark: correct evaluation. (must give exact solution)	

c)(i)	$4y = x^2 - 12$ $x^2 = 4y + 12$ $x^2 = 4(y + 3)$ $4a = 4$ $a = 1$ Focal length = 1	1 mark: gives correct focal length from correct working	
c)(ii)	Vertex $(0, -3)$ Hence, focus = $(0, -3 + 1)$ = $(0, -2)$	1 mark: uses their focal length and vertex to calculate the focus	
d)	$y = \frac{1}{2} \sin x$ at the point $(\pi, 0)$ $\frac{dy}{dx} = \frac{1}{2} \cos x$ At $x = \pi$, $\frac{dy}{dx} = \frac{1}{2} \cos \pi = -\frac{1}{2}$ Equation of the tangent is $y - 0 = -\frac{1}{2}(x - \pi)$ $2y = -x + \pi$ $x + 2y - \pi = 0$	2 marks: correct equation of the tangent from correct working 1 mark: calculates the correct gradient and attempts to find the equation of the tangent using point-gradient formula.	

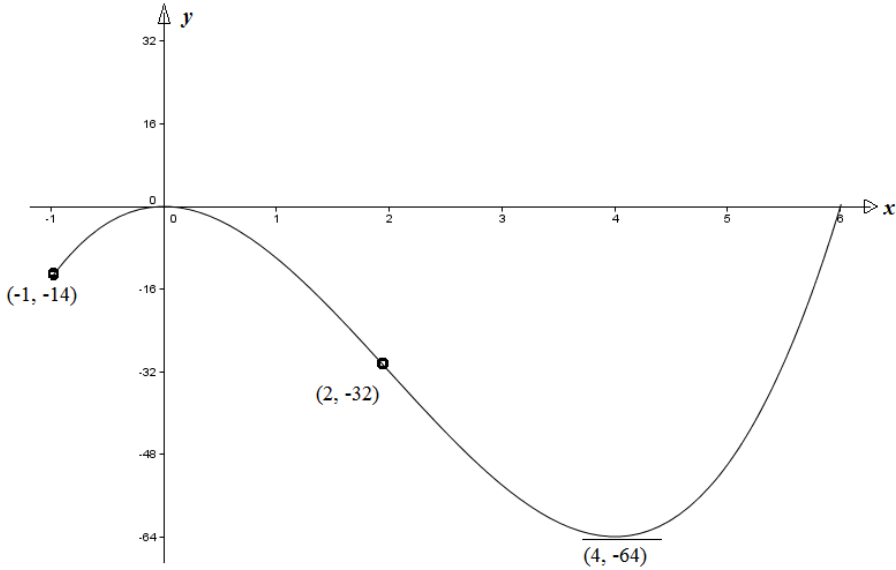
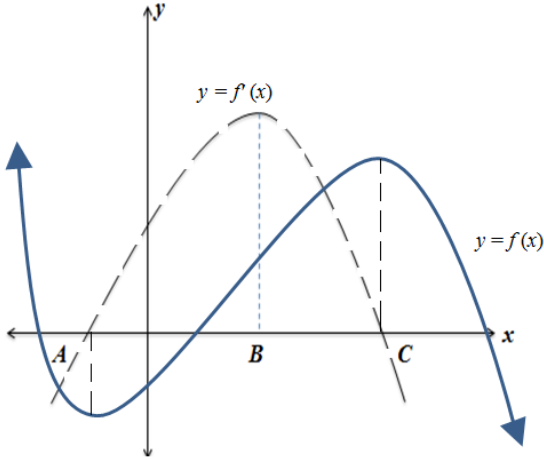
Question 13

13a)	$px^2 - x + q = 0 \quad \alpha = -2 \text{ and } \beta = 5$ $\therefore -2 + 5 = \frac{1}{p}$ $\therefore p = \frac{1}{3}$ and $-10 = \frac{q}{\frac{1}{3}}$ $\therefore q = -\frac{10}{3}$	2 marks correct values of p and q showing full working 1 mark correct value one of p or q showing full working	
13bi)	$m_{BC} = \frac{5-2}{1-0} = 3$ line AD has equation $y - 3x + 1 = 0$ $\therefore y = 3x - 1$ $\therefore m_{AD} = 3$ and $m_{AB} = \text{undefined (vertical)}$ and $m_{CD} = 0$ (horizontal) $\therefore ABCD$ is a trapezium as it has one pair of parallel sides	2 marks Concludes shape is a trapezium by comparing all 4 gradients 1 mark only compares gradients of parallel sides	
13bii)	Point D is the intersection of $y = 5$ and $y - 3x + 1 = 0$ $\therefore 5 - 3x + 1 = 0$ $\therefore x = 2$ and $y = 5$ (horizontal line) $D(2, 5)$	1 mark Finds coordinates of D using any correct method	
13biii)	$l_{BC} = \sqrt{(1-0)^2 + (5-2)^2}$ $= \sqrt{10}$	1 mark correctly evaluates the length of BC	
13biii)	$B(0, 2) \quad -3x + y + 1 = 0$ $\perp_{dist} = \frac{ (-3 \times 0) + (1 \times 2) + 1 }{\sqrt{(-3)^2 + 1^2}}$ $= \frac{3}{\sqrt{10}}$	2 marks correct substitution into perpendicular distance formula 1 mark – minor error	
13biv)	$l_{AD} = \sqrt{(2-0)^2 + (5-(-1))^2}$ $= \sqrt{40} = 2\sqrt{10}$ $A_{ABCD} = \frac{1}{2}(l_{AD} + l_{BC}) \times \frac{3}{\sqrt{10}}$ $= \frac{3\sqrt{10}}{2} \times \frac{3}{\sqrt{10}}$ $= \frac{9}{2} \text{ units}^2$	2 marks any correct method for finding area of trapezium using their point D 1 mark significant progress towards finding area including correct formula for area of a trapezium	

c)(i)	$\angle RSP = \angle PQR = \alpha$ (opposite angles of a parallelogram are equal) $\therefore \angle RSU = \angle TQR = 180 - \alpha$ (supplementary angles add to 180°) In $\triangle RQT = \triangle USR$, $\angle TQR = \angle RSU$ 1. $QR = SU$ ($QR = PS$ (opposite sides of a parallelogram are equal). $PS = SU$ (given) 2. $PU \parallel QR$, opposite sides of a parallelogram are parallel) $\angle QRT = \angle SUR$ ($PU \parallel QR$, corresponding angles in parallel lines are equal) 3. $\therefore \triangle TQR \equiv \triangle RSU$ (angle- angle- side) using 1., 2., 3.	3 marks: correct proof given 2 marks: proves $\therefore \angle RSU = \angle TQR$ and $\angle QRT = \angle SUR$ with correct reasoning 1 mark: anyone of 1., 2. or 3. proved with correct reasoning	
c)(ii)	$\therefore SR = QT$ corresponding sides of congruent triangles are equal 1 mark $QT = PQ$ (given) $\therefore PQ = SR$ Now, $PS = QR$, $PQ = SR$ (all sides are equal, $PQRS$ is a rhombus 1 mark		

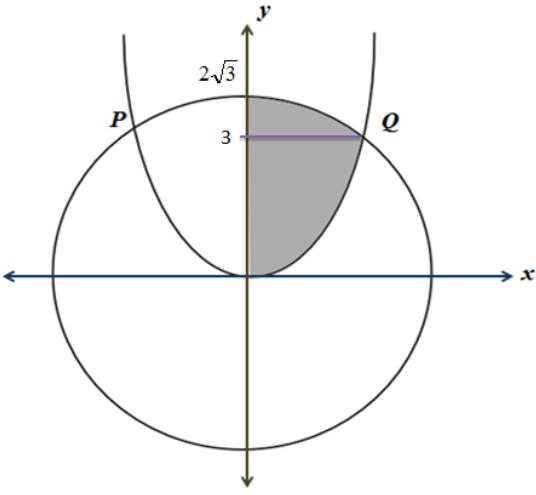
Question 14

a)(i)	$f(x) = 2x^3 - 12x^2$ For stationary points, $f'(x) = 0$ $6x^2 - 24x = 0$ $6x(x - 4) = 0$ $x = 0, \quad x = 4$ $x = 0, \quad x = 4$ $y = 0 \quad y = -64 \quad \text{1 mark}$ $(0,0) \quad (4,-64)$ Determining nature: $f''(x) = 12x - 24$ $x = 0, \quad f''(x) = -24 < 0$ $\therefore$ maximum turning point at $(0,0)$ 1 mark $x = 4, \quad f''(x) = 24 > 0$ $\therefore$ minimum turning point at $(4,-64)$ 1 mark	1 mark: finds the two stationary points 1 mark each: for determining the nature of each of the stationary points													
(ii)	For points of inflection, $f''(x) = 0$ $12x - 24 = 0$ $x = 2$ $x = 2, \quad y = -32$ Concavity test: <table border="1" data-bbox="239 1523 718 1702"> <thead> <tr> <th>x</th><th>1</th><th>2</th><th>3</th></tr> </thead> <tbody> <tr> <td>$f''(x)$</td><td>-12</td><td>0</td><td>12</td></tr> <tr> <td></td><td></td><td></td><td></td></tr> </tbody> </table> Change in concavity. $\therefore$ point of inflection at $(2,-32)$	x	1	2	3	$f''(x)$	-12	0	12					1 mark: sets $f''(x) = 0$ and finds the possible point of inflection 1 mark: performs concavity test to establish point of inflection	
x	1	2	3												
$f''(x)$	-12	0	12												
															

(iii)		1 mark: correct shape 1 mark: Correct end points clearly shown and labelled 1 mark: their stationary points and point of inflection labelled.
b)(i)	Minimum turning point at A and point of inflection at B. 	1 mark for each
(ii)	The function is decreasing when $f'(x) < 0$. $\therefore x < A$ and $x > C$	1 mark (need both regions to get the mark)
b)(iii)		1 mark: correct shape 1 mark: positions of A, B and C clearly marked and correct nature for each of the points demonstrated

	$\int_{-1}^1 4^x dx$ using Simpsons rule with 5 function values					2 marks – correctly applies Simpsons rule with 5 function values 1 mark – has table of values but small calculation error or correctly states Simpsons rule but error in table of values
	x	-1	-0.5	0	0.5	
	y	0.25	0.5	1	2	
	$\int_{-1}^1 4^x dx \approx \frac{0.5}{3}(0.25 + 4 \times 0.5 + 2 \times 1 + 4 \times 2 + 4)$ $= \frac{65}{24}$					

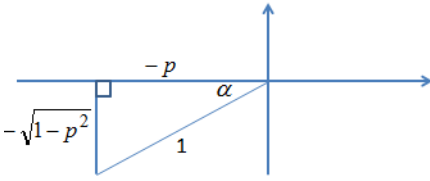
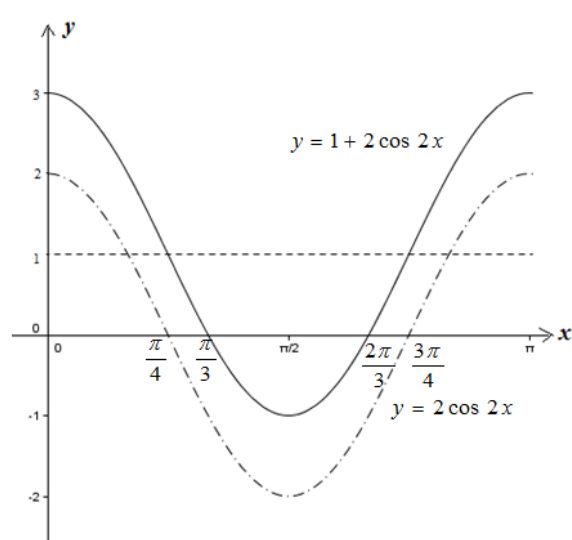
Question 15

a)	$A = \int_0^2 ((-2x) - (x^2 - 4x)) dx$ $= \int_0^2 (2x - x^2) dx$ $= \left[x^2 - \frac{x^3}{3} \right]_0^2$ $= \left[4 - \frac{8}{3} \right] - 0$ $= \frac{4}{3} \text{ units}^2$	1 mark1 correctly identifies the integral to be evaluated between the two curves - must have functions in correct order 1 mark finds correct primitive for their integral 1 mark correctly substitutes bounds for their integral (if student just takes absolute value at end then they lose 1 mark)
b)(i)	$y = x^2$ and $x^2 + y^2 = 12$.  $x^2 + y^2 = 12$. $y = x^2$ Solving simultaneously, $x^2 + x^4 - 12 = 0$ $x^4 + x^2 - 12 = 0$ $(x^2 + 4)(x^2 - 3) = 0 \quad \text{1 mark}$ $x^2 + 4 = 0$, No solution $x^2 - 3 = 0 \quad \therefore x = \pm\sqrt{3}$ Then $y = (\sqrt{3})^2 = 3 \quad \text{1 mark}$	1 mark: writes the simultaneous equation and factorises correctly. 1 mark: correctly solves for x and hence proves the result (MUST reject the $x^2 + 4 = 0$)

(ii)	$V = \pi \int x^2 dy$ $y = x^2$ $y^2 = 12 - x^2$ $= \pi \int_0^3 y dy + \pi \int_3^{2\sqrt{3}} (12 - y^2) dy$ $= \pi \left[\frac{y^2}{2} \right]_0^3 + \pi \left[12y - \frac{y^3}{3} \right]_3^{2\sqrt{3}}$ $= \pi \left[\frac{9}{2} - 0 \right] + \pi [24\sqrt{3} - 8\sqrt{3} - 36 + 9]$ $= \pi \left(16\sqrt{3} - \frac{45}{2} \right) \text{ unit}^3.$	3 marks: correct solution from correct working. 2 marks: gives the correct expression for volume and makes significant progress towards the answer 1 mark: calculates one of the volumes correctly (possibly $\frac{9}{2}\pi$
c)(i)	$x - 2y + 8 = 0$ $4x - 6y + 27 = 0$ Equation of the line that passes through the point of intersection is of the form $x - 2y + 8 + k(4x - 6y + 27) = 0 \quad \mathbf{1 \text{ mark}}$ $x(1 + 4k) - y(2 + 6k) + 8 + 27k = 0$ Gradient of this line is $m = \frac{1 + 4k}{2 + 6k} = -\frac{3}{2}$ $2 + 8k = -6 - 18k$ $26k = -8$ $k = \frac{-8}{26} = \frac{-4}{13} \quad \mathbf{1 \text{ mark}}$ Hence, the required line is $x - 2y + 8 + \frac{-4}{13}(4x - 6y + 27) = 0$ $13x - 26y + 104 - 16x + 24y - 274 = 0$ $-3x - 2y - 170 = 0$ $3x + 2y + 170 = 0 \quad \mathbf{1 \text{ mark}}$	1 mark: writes the general form of the line that passes through the point of intersection of two other lines. 1 mark: Equates the gradient of the line to $-\frac{3}{2}$ and evaluates k. 1 mark: substitutes the value of k into the general form and simplifies to give the equation

d(i)	$\frac{d}{dx} \left(\sqrt{1+x^2} \right) = \frac{1}{2\sqrt{1+x^2}} \times 2x = \frac{x}{\sqrt{1+x^2}}$	1 mark: correctly differentiates, showing all the working
d(ii)	From (i), using the anti- derivative, $\int \frac{x}{\sqrt{1+x^2}} dx = \sqrt{1+x^2} + C \quad \mathbf{1 \text{ mark}}$ $\int \frac{x}{2\sqrt{1+x^2}} dx = \frac{1}{2} \sqrt{1+x^2} + C$ $\int_0^1 \frac{x}{2\sqrt{1+x^2}} dx = \frac{1}{2} \left[\sqrt{1+x^2} \right]_0^1 \quad \mathbf{1 \text{ mark}}$ $= \frac{1}{2} (\sqrt{2} - 1) \quad \mathbf{1 \text{ mark}}$	1 mark: writes the anti-derivative statement 1 mark: gives the correct expression for evaluating the integral 1 mark: correct solution

Question 16

a)	 $\sec(90 + \alpha) = \frac{1}{\cos(90 + \alpha)}$ $= \frac{1}{-\sin \alpha} \quad \text{1 mark}$ $= \frac{1}{--\sqrt{1-p^2}}$ $= \frac{1}{\sqrt{1-p^2}} \quad \text{1 mark}$	2 marks: correct answer from correct working 1 mark: • Draws the correct diagram to show the dimensions OR • Writes $\sec(90 + \alpha) = \frac{1}{-\sin \alpha}$
b)(i)	$1 + 2 \cos 2x = 0$ $\cos 2x = -\frac{1}{2}$ $2x = \frac{2\pi}{3} \text{ or } \frac{4\pi}{3}$ $\therefore x = \frac{\pi}{3} \text{ or } \frac{2\pi}{3}$	1 mark: solves the equation or shows by substitution
(ii)		1 mark: correct shape 1 mark: correct intercepts and shows how $y = 2 \cos 2x$ transports to $y = 1 + 2 \cos 2x$ at $\frac{\pi}{4}$ and $\frac{3\pi}{4}$ at $y = 1$ 1 mark: an even scale in terms of $\frac{\pi}{4}$ and $\frac{\pi}{3}$ are drawn on a rules axis

(iii)	$\text{Area} = \left \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} 1 + 2 \cos 2x \, dx \right $ $= \left \left[x + \sin 2x \right]_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \right $ $\left \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) - \left(\frac{\pi}{3} + \frac{\sqrt{3}}{2} \right) \right $ $= \left \left(\frac{\pi}{3} - \sqrt{3} \right) \right $ $\sqrt{3} - \frac{\pi}{3} \text{ sq. units}$	2 marks: correct answer from correct working 1 mark: incorrect evaluation or not using the absolute for area. (Take a mark off, if inconsistent with absolute sign from line to line.)
c)(i)	Perimeter $P = r\theta + 2r$ $r\theta + 2r = 375$ 1 mark $\theta = \frac{375 - 2r}{r}$ $A = \frac{1}{2} r^2 \theta$ $= \frac{1}{2} r^2 \cdot \frac{375 - 2r}{r}$ $= \frac{r}{2} (375 - 2r)$	1 mark: gives the correct expression for perimeter 1 mark: correctly derives the result showing all the working
(ii)	From (i), $A = \frac{375}{2} r - r^2$ $\frac{dA}{dr} = \frac{375}{2} - 2r = 0$ $2r = \frac{375}{2}$ $r = \frac{375}{4} \quad \textbf{1 mark}$ $\frac{d^2 A}{dr^2} = -2 < 0$ $\therefore$ Area is maximum when $r = \frac{375}{4}$ 1 mark $\text{Maximum area} = \frac{1}{2} \times \frac{375}{4} \left(375 - 2 \times \frac{375}{4} \right) \quad \textbf{1 mark}$ $= 8789.0625 \text{ m}^2$	1 mark: finds the stationary point correctly 1 mark: proves that the area is maximum 1 mark: finds the maximum area

(iii)	From (i), $\theta = \frac{375}{r} - 2$ $= \frac{375}{\left(\frac{375}{4}\right)} - 2$ $= 2 \text{ radians}$ $= 2 \times \frac{180}{\pi}$ $= 114^{\circ}35'$ $110^{\circ} < 114^{\circ}35'$ No, the built area will be less	2 mark: correct answer with correct working. (MUST state $110^{\circ} < 114^{\circ}35'$) 1 mark: calculates θ and makes some reasonable comment based on their value of θ.
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